

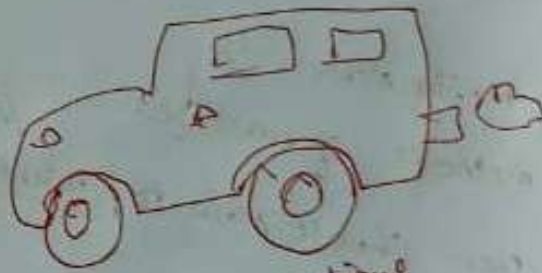
objects



chair



or any object



Some actions

Exposed to user

Some are hidden

Exposed to user:

How to Drive
Where to put engine
Wheel motion

Hidden

Engine runs the car
How petrol consumed

As long as car satisfies

we don't care



When you
at tree
is ob

But with
imagined
defin

Some

Imo

Encar

Class & Objects



When you look
at tree It
is object

But what you
imagine way you
define it is as class

way you define

→ Template

What actual thing

→ object

Sometimes it specifies how and what
things are done.

Imagine a Object what and how it
does → Defines (Class)

Object → actual instance of it

Encapsulation



Car owner does need to know
How engine makes the car
run.

(Information Hiding)

Encapsulation



Someone
doesn't
want you
to know
how they
do things

But they
want
you to
use the
products

Hide Implementations
Expose Necessary

Object

what it has



instance fields

what it does



Methods

Set of those
values



Current state



black box

No one
wants to
know how
you implement

Objects

Everything

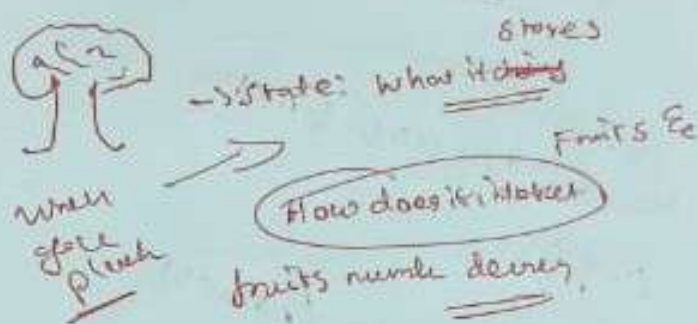
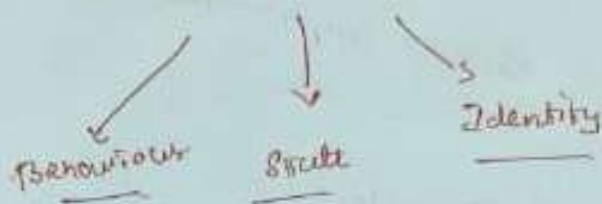
Behaviour



when
you
push

objects

Everything we need in world is objects



Behaviour $\rightarrow$ Action (pluck fruit)

Identity $\rightarrow$ State & Behaviour



Distinguishes
same variety of trees

Behaviour $\rightarrow$ method you call

information $\rightarrow$ this time's state

object state change over time by series of method

like a tree withers over series of actions.

new Date();

defined

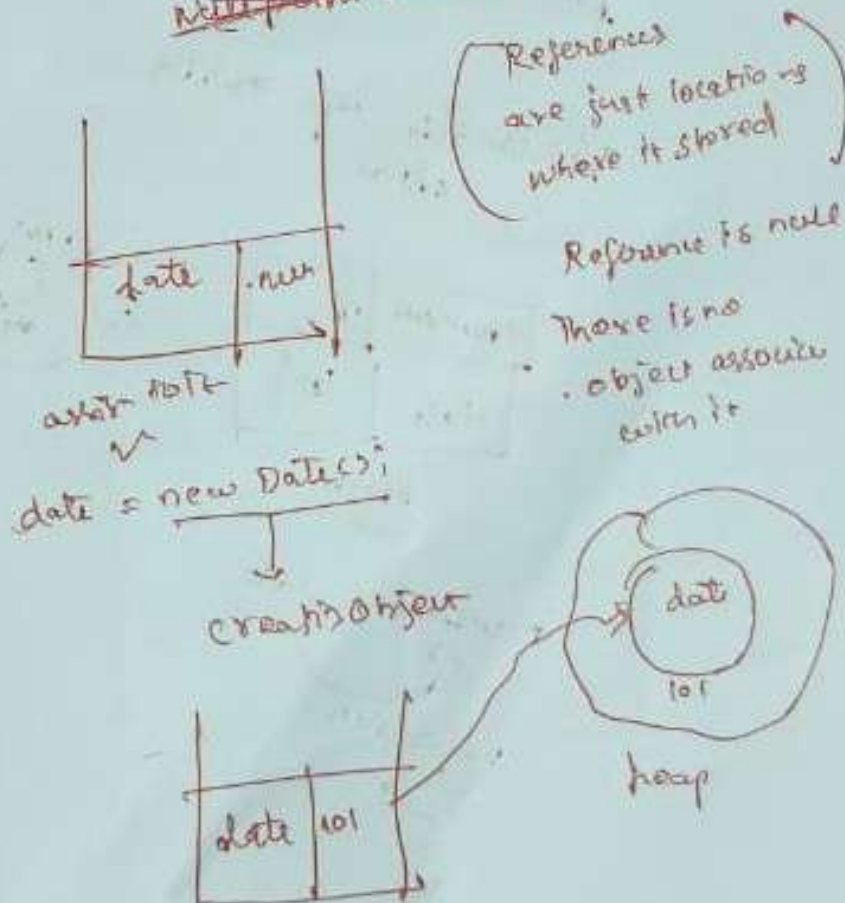
new $\rightarrow$ returns the reference
of class

Date(): Constructor
~~then~~ invokes
Object creates

Date date; $\rightarrow$ You initialize
object variable

at this point there is no object
associated with it

date.toString() $\times$ Compile-time
error
~~Null pointer exception~~



object's class → without
metho

Someone compromised

Identifying class

Identify class & metho

oops

look for noun (names)

metho. & → verbs (actions)

Relationship b/w
classes

is-a
has-a
is-a



another person's
A man used ~~his~~ account

A man used depends on another

(child depends on mother)

Class A $\leftarrow$ Class B

Class A depends on class B

for eg:-

Order

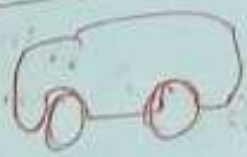
Paid Bill

Account

(Coupling)

Must reduce
coupling

has-a relation



Car has a tyre

Aggregation

is-a relation (Advantage)

generalise
↓

Superman is a man



specialised
version of man

Predefined
class

create-already defined (pre-defined)

Math class → (random)

You don't know
or you don't care
How it is implemented

Encapsulation

Math → only hides implementation

Date → point of time

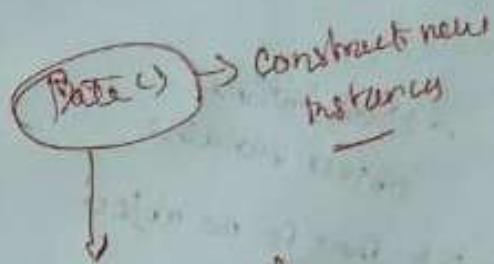
Date date; → object
variable



It does not associate
with any object
yet -

new -> return reference
and stored in stack

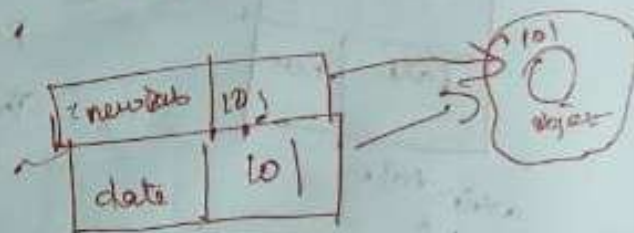
Reference Type where actual
value is stored //



Same name as
class name

`Date date = new Date();`

You can also refer existing
object

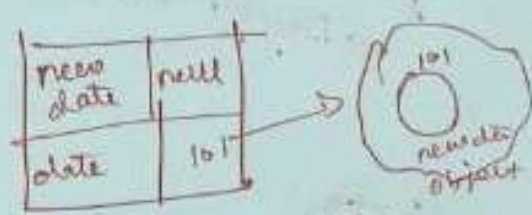


Stack

`newDate = date`
reference

Object variable
↓
references to object

~~make~~ date = null; ✗ makes no object associated with it



Local date class

Date class → point of particular instance of time (epoch)

Calendar (Gregorian Calendar)

Local Date → for human use

Local Date date = LocalDate.now();
↓
creates an object

using factory methods → construct object

Local Date of (1991, 12, 31);



creates object

To find out

get Year()

get Month Value()

get Day of month()

~~get hour and days later()~~



new Local
Date object

New Year Eve plus days (1000);



Why

Date	Local Date
Port of time	Calendar

why we need separate class
for it?

Different calendars } → Different
Different format } countries
